

Electronics Device and Circuit (EDC)

Past Paper Solutions Manual

Compiled from IOE Exams

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Contents

1 BJT Biasing and Amplifiers

2083 Baishakh, Q1

Why BJT is bipolar and FET is unipolar? Design β -independent type DC biased CC amplifier ($V_{cc} = 20V, I_c = 2mA, \beta = 100$) using firm biasing.

Solution:

Bipolar vs Unipolar: A BJT (Bipolar Junction Transistor) is called bipolar because its operation depends on the flow of both majority and minority charge carriers (electrons and holes). A FET (Field Effect Transistor) is called unipolar because its operation depends only on the flow of majority carriers (either electrons in an n-channel or holes in a p-channel FET).

Design of CC Amplifier (Firm Biasing): Given: $V_{CC} = 20V, I_C \approx I_E = 2mA, \beta = 100$. For a Common Collector (CC) amplifier (Emitter Follower), we typically set $V_E = V_{CC}/2 = 10V$ for maximum symmetrical swing. $R_E = \frac{V_E}{I_E} = \frac{10V}{2mA} = 5k\Omega$.

For Firm Biasing, the rule of thumb is: $I_{R1} = 10 \times I_B$, or $R_B \leq 0.1 \times \beta R_E$. Let's use the firm bias rule: $R_{th} \leq 0.1\beta R_E = 0.1(100)(5k\Omega) = 50k\Omega$. Let $R_{th} = 50k\Omega$. $V_B = V_E + V_{BE} = 10V + 0.7V = 10.7V$. We know $V_B = \frac{R_2}{R_1 + R_2} V_{CC}$ and $R_{th} = R_1 || R_2 = \frac{R_1 R_2}{R_1 + R_2}$. Thus, $V_B = \frac{R_{th}}{R_1} V_{CC} \implies 10.7V = \frac{50k\Omega}{R_1} \times 20V$. $R_1 = \frac{50k\Omega \times 20}{10.7} = 93.45k\Omega$. $R_{th} = \frac{R_1 R_2}{R_1 + R_2} \implies 50k = \frac{93.45k \times R_2}{93.45k + R_2}$. Solving for R_2 : $R_2 = 107.5k\Omega$. Use standard values: $R_1 = 91k\Omega, R_2 = 110k\Omega, R_E = 5.1k\Omega$.

2083 Baishakh, Q2

Explain the role of bypass capacitor in amplifier output circuits. Draw the small signal equivalent circuit of a common base amplifier circuit and find its input impedance, output impedance and voltage gain.

Solution:

Role of Bypass Capacitor: The emitter bypass capacitor C_E provides a low reactance path for AC signals, effectively bypassing the emitter resistor R_E to ground for AC. This prevents negative AC feedback (emitter degeneration) and ensures that the voltage gain of the amplifier is not significantly reduced.

Common Base Amplifier (Small Signal Analysis): (Draw a small-signal r_e or hybrid- π model for CB). Let r_e be the dynamic emitter resistance. Input Impedance: $Z_{in} \approx r_e || R_E \approx r_e$ (since $r_e \ll R_E$). Output Impedance: $Z_{out} = R_C$ (assuming r_o is very large). Voltage Gain: $A_v = \frac{v_{out}}{v_{in}} = \frac{I_c R_C}{I_e r_e} \approx \frac{R_C}{r_e}$. Current Gain: $A_i = \alpha \approx 1$.

2081 Ashwin, Q1

Design β independent type dc biased common emitter amplifier. Given parameters $V_{cc} = 20V, I_c = 1.5mA$ and $\beta = 110$. Use stiff biasing method.

Solution:

Given: $V_{CC} = 20V, I_C = 1.5mA, \beta = 110$. Stiff bias rule: $R_{th} \leq 0.01\beta R_E$. Set $V_E = 0.1V_{CC} = 2V$. $R_E = \frac{2V}{1.5mA} = 1.33k\Omega$. $V_B = V_E + 0.7 = 2.7V$. Let $R_{th} = 0.01(110)(1.33k) = 1.46k\Omega$. $V_B = \frac{R_{th}}{R_1} V_{CC} \implies 2.7 = \frac{1.46k}{R_1} (20) \implies R_1 = 10.8k\Omega$.

$R_2 = \frac{R_1 R_{th}}{R_1 - R_{th}} = \frac{10.8 \times 1.46}{10.8 - 1.46} = 1.69 k\Omega$. Set $V_C = \frac{V_{CC} + V_E}{2} = 11V \Rightarrow R_C = \frac{20 - 11}{1.5mA} = 6k\Omega$.
Standard values: $R_E = 1.2k\Omega$, $R_C = 6.2k\Omega$, $R_1 = 10k\Omega$, $R_2 = 1.5k\Omega$.

2 FET and MOSFETs

2083 Baishakh, Q4

Find I_D and V_{DS} for the given JFET circuit. $V_p = -4V$, $I_{DSS} = 8mA$. (Assuming $V_{DD} = 16V$, $R_1 = 2.1M\Omega$, $R_2 = 270k\Omega$, $R_D = 2.4k\Omega$, $R_S = 1.5k\Omega$).

Solution:

$V_G = \frac{R_2}{R_1 + R_2} V_{DD} = \frac{270k}{2.1M + 270k} 16V = 1.82V$. $V_{GS} = V_G - I_D R_S = 1.82 - 1500 I_D$.
Shockley's Equation: $I_D = I_{DSS} (1 - \frac{V_{GS}}{V_p})^2$. $I_D = 8 \times 10^{-3} (1 - \frac{1.82 - 1500 I_D}{-4})^2$. Let I_D be in mA: $I_D = 8 (1 + \frac{1.82 - 1.5 I_D}{4})^2 = 8 (1 + 0.455 - 0.375 I_D)^2$. $I_D = 8 (1.455 - 0.375 I_D)^2 = 8 (2.117 - 1.09 I_D + 0.14 I_D^2)$. $I_D = 16.936 - 8.72 I_D + 1.12 I_D^2$. $1.12 I_D^2 - 9.72 I_D + 16.936 = 0$.
Solving the quadratic equation: $I_D = \frac{9.72 \pm \sqrt{(9.72)^2 - 4(1.12)(16.936)}}{2(1.12)} = \frac{9.72 \pm \sqrt{94.478 - 75.87}}{2.24} = \frac{9.72 \pm 4.31}{2.24}$. $I_D = 2.41mA$ or $I_D = 6.26mA$. Check V_{GS} for both: If $I_D = 2.41mA$, $V_{GS} = 1.82 - 1.5(2.41) = -1.795V$ (Valid, $> V_p$). If $I_D = 6.26mA$, $V_{GS} = 1.82 - 1.5(6.26) = -7.57V$ (Invalid, $< V_p$). Thus, $I_D = 2.41mA$. $V_{DS} = V_{DD} - I_D (R_D + R_S) = 16 - 2.41(2.4 + 1.5) = 16 - 2.41(3.9) = 16 - 9.399 = 6.6V$. Check saturation: $V_{DS} > V_{GS} - V_p \Rightarrow 6.6 > -1.795 - (-4) = 2.205V$. (True, saturated).

2082 Bhadra, Q4

Find I_D and V_{DS} for the circuit. Given parameters are $V_T = 1V$ and $K = 0.48mA/V^2$. (Assuming $V_{DD} = 10V$, $R_1 = 10M\Omega$, $R_2 = 10M\Omega$, $R_D = 2k\Omega$, $R_S = 1k\Omega$).

Solution:

$V_G = \frac{R_2}{R_1 + R_2} V_{DD} = \frac{10}{20} 10 = 5V$. $V_{GS} = V_G - I_D R_S = 5 - 1 I_D$ (with I_D in mA).
E-MOSFET equation: $I_D = K (V_{GS} - V_T)^2$. $I_D = 0.48 (5 - I_D - 1)^2 = 0.48 (4 - I_D)^2 = 0.48 (16 - 8 I_D + I_D^2)$. $I_D = 7.68 - 3.84 I_D + 0.48 I_D^2$. $0.48 I_D^2 - 4.84 I_D + 7.68 = 0$. $I_D = \frac{4.84 \pm \sqrt{23.42 - 14.74}}{0.96} = \frac{4.84 \pm 2.94}{0.96}$. $I_D = 1.98mA$ or $I_D = 8.1mA$. If $I_D = 8.1mA$, $V_{GS} = 5 - 8.1 = -3.1V$ (Invalid, must be $> V_T = 1V$). If $I_D = 1.98mA$, $V_{GS} = 5 - 1.98 = 3.02V$ (Valid).
Thus, $I_D = 1.98mA$. $V_{DS} = V_{DD} - I_D (R_D + R_S) = 10 - 1.98(3) = 10 - 5.94 = 4.06V$.
Check saturation: $V_{DS} > V_{GS} - V_T \Rightarrow 4.06 > 3.02 - 1 = 2.02V$. (True).

3 Oscillators and Multivibrators

2083 Baishakh, Q5

What is the main difference between RC and LC oscillator circuits in terms of application? Explain the operation of Wien Bridge Oscillator circuit with the help of circuit diagram and necessary mathematical expressions.

Solution:

Difference between RC and LC Oscillators: RC oscillators are typically used for generating low-frequency signals (audio frequencies, up to a few MHz) because large values of R and C are easy to integrate and they provide excellent frequency stability at low frequencies. LC oscillators are used for high-frequency applications (radio frequencies, MHz to GHz) because at high frequencies, the required inductor and capacitor values become small and practical to implement.

Wien Bridge Oscillator Operation: The Wien bridge oscillator consists of a two-stage RC coupled amplifier providing a phase shift of 360° (or 0°) and a Wien bridge feedback network. The feedback network consists of a series RC circuit (R_1, C_1) and a parallel RC circuit (R_2, C_2). At the resonant frequency, the phase shift introduced by the feedback network is 0° . For oscillation to occur, the Barkhausen criterion must be satisfied: total phase shift must be 0° and loop gain $\beta A \geq 1$. The feedback fraction is $\beta = \frac{V_f}{V_{out}} = \frac{Z_p}{Z_s + Z_p}$. When $R_1 = R_2 = R$ and $C_1 = C_2 = C$, the frequency of oscillation is: $f_o = \frac{1}{2\pi RC}$. At $f = f_o$, $\beta = 1/3$. Therefore, the amplifier must have a gain $A \geq 3$ to sustain oscillations.

2081 Ashwin, Q6

State Barkhausen criterion for sinusoidal oscillation. Draw the circuit diagram of Colpitts oscillator and write its frequency of oscillation.

Solution:

Barkhausen Criterion: For an oscillator to sustain steady sinusoidal oscillations, two conditions must be met: 1. The total phase shift around the closed loop must be 0° or 360° . 2. The magnitude of the loop gain must be unity, i.e., $|\beta A| = 1$ (or slightly greater than 1 to start oscillations).

Colpitts Oscillator Frequency: A Colpitts oscillator uses a tapped capacitor (C_1, C_2) and an inductor (L) in its feedback tank circuit. The equivalent capacitance is $C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$. Frequency of oscillation: $f_o = \frac{1}{2\pi\sqrt{LC_{eq}}}$.

4 Current Mirrors

2083 Baishakh, Q6

Explain how a base current compensation circuit reduces the difference between I_O and I_{REF} in a simple current mirror circuit.

Solution:

In a simple current mirror, $I_O = I_{REF} - 2I_B$. Since $I_B = I_C/\beta \approx I_O/\beta$, we have $I_O = I_{REF} \frac{\beta}{\beta+2}$. For finite β , I_O is slightly less than I_{REF} . A base current compensated current mirror introduces an additional transistor (emitter follower) to supply the base currents to the two mirror transistors. Let the third transistor be Q_3 . The emitter current of Q_3 supplies the base currents for Q_1 and Q_2 . Thus $I_{E3} = 2I_B$. The base current of Q_3 is $I_{B3} = \frac{I_{E3}}{\beta+1} = \frac{2I_B}{\beta+1}$. Now, at the input node, $I_{REF} = I_C + I_{B3} = I_O + \frac{2I_O}{\beta(\beta+1)}$. Therefore, $I_O = I_{REF} \left[\frac{1}{1 + \frac{2}{\beta^2 + \beta}} \right]$. This significantly reduces the error between I_O and I_{REF} , making the mirror less dependent on β .

2081 Ashwin, Q8

Draw a Widlar Current Source. Derive an expression for an output resistance of Widlar Current Source.

Solution:

A Widlar current source is used to generate very small constant currents using relatively small resistor values. It adds a resistor R_E in the emitter of the output transistor Q_2 . From the KVL equation around the base-emitter loops: $V_{BE1} = V_{BE2} + I_{E2}R_E$. Using $V_{BE} = V_T \ln(I_C/I_S)$, and assuming $I_C \approx I_E$, we get: $V_T \ln(I_{C1}/I_S) = V_T \ln(I_{C2}/I_S) + I_{C2}R_E$. $V_T \ln(I_{C1}/I_{C2}) = I_{C2}R_E$. Output Resistance R_o : The output resistance is significantly enhanced compared to a simple current mirror due to the emitter degeneration resistor R_E . $R_o \approx r_{o2}(1 + g_{m2}(R_E || r_{\pi 2}))$.

5 Power Amplifiers and Tuned Amplifiers

2083 Baishakh, Q7

For a class B amplifier providing a 20V peak signal to a 16Ω load (speaker) and a power supply of $V_{cc} = 30V$, determine the input power, output power, and circuit efficiency.

Solution:

Given: $V_p = 20V$, $R_L = 16\Omega$, $V_{CC} = 30V$. Output Power (AC power to load): $P_{out} = \frac{V_p^2}{2R_L} = \frac{(20)^2}{2(16)} = \frac{400}{32} = 12.5W$. Peak load current: $I_p = \frac{V_p}{R_L} = \frac{20}{16} = 1.25A$. Input Power (DC power from supply): For Class B push-pull, average current from supply is $I_{DC} = \frac{2}{\pi}I_p$. $P_{in} = V_{CC}I_{DC} = V_{CC} \left(\frac{2}{\pi}I_p\right) = 30 \left(\frac{2}{\pi} \times 1.25\right) = \frac{75}{\pi} \approx 23.87W$. Circuit Efficiency (η): $\eta = \frac{P_{out}}{P_{in}} = \frac{12.5}{23.87} \times 100\% = 52.36\%$.

2082 Bhadra, Q8

Define crossover distortion in a class B amplifier and explain how class AB operation can be used to eliminate it.

Solution:

Crossover Distortion: In a Class B push-pull amplifier, each transistor conducts for exactly half of the input cycle. However, a BJT requires a minimum base-emitter voltage (e.g., $0.7V$) to turn on. When the input signal crosses zero, neither transistor is fully conducting, resulting in a "dead zone" where the output remains zero while the input is between $-0.7V$ and $+0.7V$. This causes the output waveform to be distorted near the zero-crossing points, known as crossover distortion.

Class AB Operation: To eliminate crossover distortion, both transistors are given a small forward bias (usually using diodes or a V_{BE} multiplier). This ensures that the transistors are slightly conducting even when there is no input signal (idling current). When the input signal crosses zero, one transistor smoothly takes over from the other without a dead zone. This mode of operation, where conduction angle is slightly more than 180° , is called Class AB.

6 Voltage Regulators

2083 Baishakh, Q9

Derive the expression for stability factor of a discrete-transistor series regulator.

Solution:

The stability factor S_V (line regulation) of a voltage regulator is defined as the change in output voltage with respect to a change in input voltage, i.e., $S_V = \frac{\Delta V_o}{\Delta V_i}$. For a basic series pass regulator, the output voltage is $V_o = V_Z - V_{BE}$. If the input voltage V_i changes, the current through the Zener diode changes by $\Delta I_Z = \frac{\Delta V_i}{R}$. This causes a small change in Zener voltage: $\Delta V_Z = \Delta I_Z \cdot r_z = \Delta V_i \frac{r_z}{R}$, where r_z is the dynamic resistance of the Zener diode. Since $V_o = V_Z - V_{BE}$, assuming V_{BE} is constant, $\Delta V_o = \Delta V_Z$. Thus, $S_V = \frac{\Delta V_o}{\Delta V_i} = \frac{r_z}{R+r_z} \approx \frac{r_z}{R}$. A smaller S_V indicates a better regulator.

2083 Baishakh, Q10

Design a variable DC voltage regulator using the LM317 to output (3-15) volts.

Solution:

The output voltage of an LM317 regulator is given by: $V_{out} = V_{REF} \left(1 + \frac{R_2}{R_1}\right) + I_{ADJ} R_2$. Where $V_{REF} = 1.25V$ typically, and I_{ADJ} is very small ($100\mu A$) and can often be ignored for design approximation. Assume $R_1 = 240\Omega$. $V_{out} = 1.25 \left(1 + \frac{R_2}{240}\right)$. For $V_{out,min} = 3V$: $3 = 1.25 \left(1 + \frac{R_{2,min}}{240}\right) \Rightarrow \frac{3}{1.25} = 2.4 \Rightarrow 1 + \frac{R_{2,min}}{240} = 2.4 \Rightarrow R_{2,min} = 240 \times 1.4 = 336\Omega$. For $V_{out,max} = 15V$: $15 = 1.25 \left(1 + \frac{R_{2,max}}{240}\right) \Rightarrow \frac{15}{1.25} = 12 \Rightarrow 1 + \frac{R_{2,max}}{240} = 12 \Rightarrow R_{2,max} = 240 \times 11 = 2640\Omega$. To design this, R_2 can be formed by a fixed resistor $R_{fixed} = 330\Omega$ in series with a potentiometer R_{pot} . When pot is 0, $R_2 = 330\Omega$, $V_{out} \approx 2.97V \approx 3V$. Max $R_2 = R_{fixed} + R_{pot} = 2640\Omega \Rightarrow R_{pot} = 2640 - 330 = 2310\Omega$. Use a standard $2.5k\Omega$ potentiometer for R_{pot} .